

Maqueta: Honest Height Provenance in Open-Data 3D Urban Reconstruction Across 118 Places

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Abstract—A 3D city model is only as truthful as its building heights, and a model built from open public data usually gets those heights not by measuring them but by inferring them, from a digital surface model, a floor count, or a blanket prior, while presenting the result as if it were surveyed. Maqueta is a multi-modal geospatial fusion tool that reconstructs real-world areas from open data (building footprints and roads from Overture, terrain from the Copernicus digital surface model, land cover from ESA WorldCover, water and green from OpenStreetMap, population from GHS-POP, and 2.5D building heights from Google Open Buildings), and its distinguishing choice is that every building carries an explicit *height provenance*: whether its height was measured, inferred from floors, derived from the raster surface model, or assumed. This report reports that provenance across the 118 baked places, and the result is a useful corrective to how open 3D city data is usually presented: only 8.8% of building heights are authoritative measurements, while 73.2% are raster-derived, 11.6% are blanket priors, and 6.4% are floor-count inferences, so more than nine in ten heights in an open-data city model are inferred, not measured. Where authoritative LoD2 ground truth exists to check against, the fused heights carry a building-height root-mean-square error of 3 to 8 metres with a downward bias, but such ground truth is itself scarce: only two of the 118 places have public LoD2 models to validate against. Making the provenance explicit per building, rather than laundering inference into apparent measurement, is the report’s contribution; every number and figure regenerates from the committed benchmark.

Index Terms—3D city models, geospatial data fusion, building height, LoD2, open data, provenance, digital surface model, reproducibility.

I. INTRODUCTION

A 3D city model is used for solar-potential studies, flood and wind simulation, visibility and noise analysis, and urban planning [1], and every one of those uses depends on the building heights being right. Authoritative models at level-of-detail 2 (LoD2), reconstructed from airborne lidar, exist for a handful of well-mapped countries [3], but for most of the world the heights are not surveyed; they are *inferred*, from a digital surface model, from a mapped floor count, or from a blanket assumption, and the literature is clear that generating 3D city models without elevation data is a matter of estimation, not measurement [2]. The problem in practice is not that heights are inferred, which is unavoidable at global scale, but

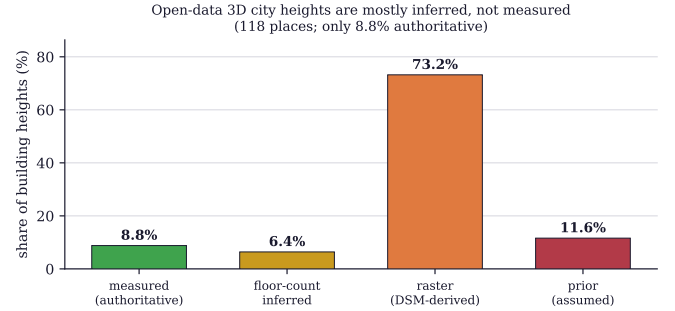


Fig. 1. The provenance of building heights across the 118 fused places. Only 8.8% are authoritative measurements; 73.2% are derived from the raster digital surface model, 11.6% are blanket priors, and 6.4% are inferred from a mapped floor count. More than nine in ten heights in an open-data city model are inferred rather than measured, which the tool records per building rather than hiding.

that the inference is usually hidden: a single-source extrusion is rendered as a crisp 3D city and presented as if every height were known.

Maqueta takes the opposite stance. It is a multi-modal fusion tool that reconstructs an area from many open sources, building footprints and roads from Overture Maps, terrain from the Copernicus GLO-30 surface model, land cover from ESA WorldCover [5], water, green and rail from OpenStreetMap, population from GHS-POP, and 2.5D building heights from Google Open Buildings [4], each layer carrying its source, licence and fetch date, and it registers them onto the height map as the 3D scaffold. Its distinguishing choice is that every building records *how* its height was obtained, measured, floor-inferred, raster-derived or assumed, so the model is honest about its own uncertainty. This report reads that provenance across the 118 baked places and benchmarks the heights where authoritative ground truth exists. The contribution is not a new reconstruction algorithm; it is the explicit, quantified provenance, and the corrective it provides to how open 3D city data is usually presented.

II. HEIGHT PROVENANCE ACROSS 118 PLACES

The core result is Fig. 1, and it is a plain statement of what an open-data 3D city model actually is. Aggregated over the 118 places and their millions of buildings, the height provenance breaks down as: 8.8% measured (an authoritative source gives the height directly), 6.4% inferred from a mapped floor count, 73.2% derived from the Copernicus raster surface

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Technical report, 2026. Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_RES_Maqueta; the *geoscena* fusion core at https://github.com/fsantibanezleal/CAOS_GeoScena.

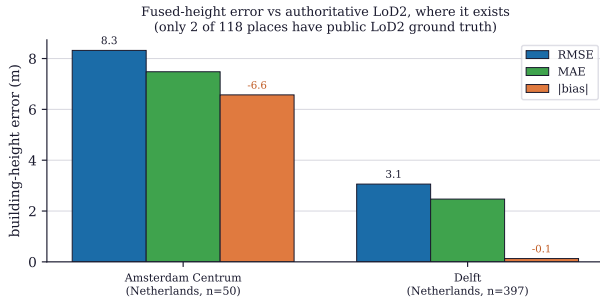


Fig. 2. Fused-height error against authoritative LoD2 ground truth, for the two of the 118 places that have a public LoD2 model (both in the Netherlands, from the 3D BAG). The root-mean-square error is 3.1 m for Delft and 8.3 m for Amsterdam, both with a downward bias (heights under-estimated). That only two places can be validated at all is itself the point: authoritative 3D ground truth is scarce, so the provenance record matters more.

model (the difference between the surface and the terrain, over the footprint), and 11.6% a blanket prior where nothing better exists. The dominant source, by far, is the raster surface model, which is an *estimate* of height, not a measurement, and which is known to carry metre-scale error and a systematic bias. The honest reading is that an open-data 3D city is overwhelmingly inferred: fewer than one building in ten has a measured height. This is not a deficiency unique to Maqueta, it is the reality of global open geodata, but Maqueta’s choice to record the provenance per building means a downstream user can filter, weight or distrust a height according to how it was obtained, rather than treating a raster inference and a survey measurement as equally solid.

III. BENCHMARK WHERE GROUND TRUTH EXISTS

Where an authoritative LoD2 model exists, the fused heights can be checked against it (Fig. 2). Two of the 118 places have a public LoD2 reference, both in the Netherlands, whose national 3D BAG reconstructs LoD2 models for all buildings from lidar [3]. Against it, Maqueta’s fused heights carry a root-mean-square error of 3.1 metres at Delft and 8.3 metres at Amsterdam, with a mean absolute error of 2.5 and 7.5 metres and a downward bias in both (the inferred heights systematically under-estimate the true ones, by 0.1 m at Delft and 6.6 m at Amsterdam). The variation between the two, Delft accurate to a few metres, Amsterdam off by more, tracks how much each relied on inference versus better sources, exactly the distinction the provenance record captures. But the more important number is the denominator: only 2 of 118 places can be validated against authoritative ground truth at all, because LoD2 models are published for so few places. That scarcity is why the provenance matters: for the other 116 places there is no ground truth to appeal to, and the honest thing a model can do is say, per building, how confident it should be.

IV. DISCUSSION AND SCOPE

Maqueta’s value is honesty about a fact the field usually elides: an open-data 3D city model is mostly inferred, and the tool makes that inference explicit and auditable per building rather than rendering it as crisp truth. The fusion itself is real

and multi-modal, eight open layers registered onto a common height scaffold with full source-and-licence provenance, and the benchmark, where it can run, quantifies the metre-scale height error. The scope is stated plainly. The heights are inferred for over ninety percent of buildings, and the report says so; the two LoD2 benchmarks are a thin validation set, not because the tool was tested lightly but because authoritative 3D ground truth genuinely exists for almost nowhere, which is itself a finding. The reconstructed scenes are for visualisation, teaching and planning context, not survey-grade models, and each carries an honest height-provenance and source manifest so no layer is mistaken for more than it is. The reusable fusion core, geoscena, turns any area of interest into a provenance-tracked scene bundle, and the product pipeline bakes a curated registry of places on top of it. None of this is hidden; the interface shows the provenance of every selected building.

V. CONCLUSION

Maqueta reconstructs 3D scenes of 118 real places by fusing eight open geospatial modalities, and its honest height provenance shows what such a model is made of: only 8.8% of building heights are authoritative measurements, the rest inferred from a raster surface model, a floor count or a prior. Where authoritative LoD2 ground truth exists, at just two of the 118 places, the fused heights carry a 3-to-8-metre error with a downward bias. Recording that provenance per building, and reporting how scarce ground truth actually is, rather than laundering inference into apparent measurement, is the report’s discipline, and the reproducible fusion tool that does it is its contribution.

APPENDIX A REPRODUCTION

The committed benchmark (data/derived/benchmark.json) carries, per place, the layer counts, triangle and size totals, the height-provenance mix (measured / floors / raster / prior) and, where a LoD2 reference exists, the fused-versus-truth root-mean-square error, mean absolute error and bias. The figure scripts under manuscripts/geo-reconstruction/figures/ aggregate it into the global provenance mix and the LoD2 benchmark. The geoscena fusion core and the maquetalab pipeline are seeded and versioned, so a place re-bakes deterministically; the open source layers are fetched under their respective licences and are not re-hosted.

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